Social Network

[ASSIGNMENT 01]

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1 Theoretical Questions

1.1 Question 1: TheWatts-Strogatz Model

1.1.1 (a) Network Construction and Simulation

This part is shown in Figure **??**

1.1.2 (b) Clustering Coefficient when (p ¿ 0)

1. Compare your simulation results with the theoretical predictions.
The Watts-Strogatz model starts with a regular ring lattice. In this regular lattice, when $p = 0$, each node is connected to its $\frac{\langle k \rangle}{2}$ neighbors on either side.

Coefficient in a Regular Lattice:
In a regular ring lattice, the clustering coefficient for a node i is defined as the ratio of the number of triangles connected to i over the maximum number of triangles that could be formed by i's neighbors. In a regular ring lattice:
Each node has exactly $\langle k \rangle$ neighbors $degree\langle k \rangle$.
These neighbors are organized in a cycle (a ring), meaning each node is connected to two neighbors, and their neighbors are similarly connected.
For a regular lattice, the number of triangles that can be formed between a node and its neighbors is related to the number of connected neighbors. For this case, the clustering coefficient C_0 for the regular lattice before rewiring, $p = 0$ is:
$C_0 = \frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)}$
$\langle k \rangle$ is the average degree of the nodes in the graph. This term comes from the structure of the regular lattice, where the nodes have a specific pattern of interconnections that result in a certain number of triangles. This is derived from calculating the number of triangles in a regular lattice and then dividing by the total number of potential triangles that can be formed given the degree of nodes.
Thus, when $p = 0$, the clustering coefficient $C(0)$ of the graph is:
$C(0) = \frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)}$

2. Explain why the Random Network follows a logarithmic scaling while lattices follow a polynomial scaling.
When $p > 0$, some of the edges are rewired randomly. This random rewiring introduces randomness into the graph and reduces the clustering coefficient, as some of the triangles formed in the regular lattice are destroyed by the rewiring process.
The clustering coefficient $C(p)$ for $p > 0$ can be approximated as:

$$C(p) \approx \left(\frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)} \right) (1 - p)^3$$

Explanation of the Formula:

$$\frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)}$$

is the clustering coefficient for the regular lattice $p = 0$.

$(1-p)^3$ represents the reduction in clustering due to the rewiring. As the probability p increases, the randomness in the graph increases, which decreases the clustering coefficient. The factor $(1-p)^3$ captures this decay in clustering as a result of rewiring edges.
Thus, when ($p > 0$), the clustering coefficient $C(p)$ decreases with increasing rewiring probability, and this approximation reflects the fact that more random edges lower the local clustering.

For $p = 0$: The clustering coefficient is given by:

$C(0) = \frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)}$

For $p > 0$: The clustering coefficient is approximately:

$C(p) \approx \left(\frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)} \right) (1-p)^3$

1.2 Question 2: Snobbish Network

1.2.1 (a) Minimal Values for p and q for Global Connectivity

For a snobbish network, we have two communities, with 9 red nodes and N blue nodes, and two connection probabilities:

p for links between nodes of the same color,
q for links between nodes of different colors.

The network is snobbish if $p > q$, meaning nodes are more likely to connect to others of the same color. For global connectivity in the network, we need a giant connected component that includes both red and blue nodes. This requires sufficient intra-community and inter-community links.
For global connectivity, the following conditions must hold:
Each node should be connected to at least one other node from the same community ensuring internal connectivity within each color. There must be enough links between the two communities cross-links to ensure that nodes in different communities can connect indirectly or directly.
For large values of N , the minimal values of p and q ensuring global connectivity are typically derived based on the percolation theory of networks. The key idea is that each node needs a sufficient number of connections to be reachable from other nodes, and enough inter-community links must exist to bridge the communities.
Minimal Value for p (Intra-community connection probability):
For intra-community connections (within each community of (N) nodes), each node must have at least one connection to ensure no isolated subgraphs. In a large snobbish network, the minimal value for p can be approximated as proportional to the inverse of the community size:

$p_{min} \sim \frac{\log N}{N}$

This ensures that there are enough connections within each community to avoid isolated nodes.
Minimal Value for q (Inter-community connection probability):
To connect the two communities (ensure global connectivity), there must be a sufficient number of cross-links (edges between red and blue nodes). The minimal value for q can also be derived from the requirement that the inter-community links are enough to bridge between communities:

$q_{min} \sim \frac{\log N}{N}$

This ensures that there are enough cross-community edges to maintain global connectivity.
Thus, the minimal values for p and q are:

$p_{min} \sim \frac{\log N}{N}, \quad q_{min} \sim \frac{\log N}{N}$

1.2.2 (b) Scaling of Average Shortest Path Length

1. Expected Scaling of $\langle d_{same} \rangle$ within the same color:
When $p \gg q$, the network is snobbish, meaning that nodes within the same community (red or blue) are more likely to connect to each other. For large N, the average shortest path length $\langle d_{same} \rangle$ between two nodes of the same color will scale similarly to a random graph with the probabilit p . The expected scaling of the path length within each community is:

$\langle d_{same} \rangle \sim \frac{\log N}{\log p}$

This implies that as N increases, the average shortest path length grows logarithmically, but the rate of increase depends on the value of p.

2. Expected Scaling of $\langle d_{diff} \rangle$ (between different colors):
When $p \gg q$, nodes of different colors (red and blue) are less likely to connect directly due to the small probability q . However, the global connectivity criteria from part (a) ensure that there are enough inter-community links to connect nodes across communities. The expected path length between nodes of different colors $\langle d_{diff} \rangle$ will scale based on the connectivity across the communities, and for large (N), it can be approximated as:

$\langle d_{diff} \rangle \sim \frac{\log N}{\log q}$

suggests that the shortest path between nodes of different colors grows logarithmically with N , and the rate of growth depends on the value of q .

3. Does this network model exhibit the small-world property?
Small-world networks are characterized by two key properties:

Short average path length: The average shortest path length grows logarithmically with the size of the network (N), i.e., $\langle d_{same} \rangle \sim \log N$ for the same color and $\langle d_{diff} \rangle \sim \log N$ for different colors.
High clustering coefficient: A significant fraction of nodes' neighbors are themselves neighbors, resulting in a relatively high clustering coefficient.
In this snobbish network model, the following can be concluded:
Path length: The average shortest path length, both within the same color $\langle d_{same} \rangle$ and between different colors $\langle d_{diff} \rangle$, grows logarithmically with N, which satisfies the first condition of small-world networks.
Clustering: Since the probability p is high for intra-community links, the network has high clustering within each color, as each node is likely to connect to other nodes in its community. Thus, the network also satisfies the second condition of small-world networks.
Therefore, the snobbish network model does exhibit the small-world property, as it satisfies both the short path length and high clustering criteria typically associated with small-world networks.

References

[1] Barabási, A.-L., *Network Science*, Cambridge University Press, 2016.